MICROSOFT x LSEG LONDON HACKATHON 2026 · TEAM 06

SPARK

Smart Portfolio Advisor for Retail Knowledge

“Your investments shouldn’t bet on the same thing your paycheck does.”

By Bipindra, Prince and Faraaz

Educational hackathon project. Historical data and paper money only. Not financial advice.

The double hit nobody warns retail investors about

150,000+

tech workers laid off in 2022

−33%

Nasdaq fall in the same year: job and savings hit together

1.2 pp

a year lost by the average fund dollar to bad timing (10 yrs to 2025)

PAIN POINTS

Concentration

Savings sit in the same industry as the paycheck

Panic-selling

People sell in drops and lock in losses

Untrustworthy AI

Chatbots invent numbers and change answers

Sources: ETF Trends (Reuters, Layoffs.fyi); Airmail; Morningstar, Mind the Gap 2026.

Use case: Priya, 29, fintech engineer (STAR)

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Situation

$100k to invest, $15k already in tech-heavy QQQ, $30k house deposit due in 5 years. Says “high risk” but sold everything in March 2020.

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Task

Build a portfolio that doesn’t double down on tech, protects the deposit and keeps her invested through a crash.

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Action

SPARK flags the risk contradiction and steps down a level, halves tech, puts $30k in Treasuries and coaches her in the 2022 drop.

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Result

Replayed from 3 Jan 2022, she falls far less than the tech-heavy comparison and the deposit is untouched. In 2023 she lags: we show that too.

Every holding opens to its reason and its sources.

Same $100k, three different clients, three portfolios



Priya, 29

Fintech engineer, London

Less tech; safe bucket for her deposit; risk contradiction flagged



Tom, 41

NHS nurse, never invested

Stable income, retires in 25 years, so can take more share risk



Fiona, 35

Offshore oil engineer, Aberdeen

Less energy; employer-share concentration flagged; kids' education in 10 years

WHAT SPARK DOES FOR EACH OF THEM

Build

diversified, away from job

Explain

every choice, sourced

Ask

plain-English Q&A

Guard

coach, don't panic-sell

Autopilot

capped, rule-bound slice

Code decides. The LLM only explains



Client story
plain English

LLM parses
to Profile JSON

Data clock
only data up to sim date

Code engine
rules, caps, every number

LLM explains
using Facts only

Number check
reject → regenerate

● Azure AI Foundry (Azure OpenAI) ● Plain Python code (Flask) ● Approved, logged to Azure Cosmos DB

Invented numbers

Every number must exist as a Fact with a source and date. Anything else is rejected before the client sees it.

Hindsight leakage

The LLM knows 2022 from training, so it never predicts. Live leak test: it refuses to say what tech does next.

Prompt injection

“Buy one meme stock” fails: trades come only from an ETF whitelist and rules live in code, not the prompt.

Not one feed. Many signals, every number traced

THE RISK

Trusting one source like Yahoo Finance

- Prices say what a fund did, not what the client's job already exposes them to.
- Adjusted prices are rewritten with later dividends, so future data leaks in.
- One feed's errors become your recommendation.

Yahoo Finance

ETF prices, plain Close (not adjusted)

FRED

Inflation and T-bill rates for the safe bucket

Kaggle layoffs

Which industries are shedding jobs

Fund factsheets

What each ETF actually holds, by sector

The client

Job, goals, holdings and past behaviour

Fact provenance

Every number tagged source + as-of date

Built for retail investors. Ready for small insurers

SPARK step	Retail investor (built)	Small insurance firm (next)
Profile	Job, goals, risk, holdings	Lines of business, claims liabilities, capital
Exposure map	Job → avoid same sectors	Underwriting book → avoid assets hit by the same event
Safe bucket	House deposit in Treasuries	Expected claims matched by short bonds
Guard	Coach through drops	Alert when a drop eats into the capital buffer
Number check + log	Trustworthy advice	Audit trail a regulator can follow

Example: a UK home insurer shouldn't also be heavy in UK property. Same idea as Priya and tech.

Why SPARK stands out

	AI chatbot	Robo-adviser	SPARK
Knows your job's exposure	✗	✗	✓
Every number checked against data	✗	partly	✓
Same answer every time, logged	✗	✓	✓
Coaches you through a crash	✗	✗	✓
Can't peek at the future in tests	✗	partly	✓



Who would use it

Banks and investing apps offering the FCA's new targeted support (since April 2026) need suggestions they can explain and record.

Don’t double down.

Your investments shouldn’t bet on the same thing your paycheck does.

Honest

Paper money, historical data; we show the 2023 lag

Compliant path

A real launch goes through an FCA-authorised partner

Next

LSEG data feeds, UK funds and ISAs, more job types